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to the
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the South Kensington
Museum

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*** START OF THE PROJECT GUTENBERG EBOOK A BRIEF GUIDE
TO THE FOOD COLLECTION ***

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SCIENCE AND ART DEPARTMENT
OF THE COMMITTEE OF COUNCIL ON EDUCATION.

BETHNAL GREEN BRANCH OF THE
SOUTH KENSINGTON MUSEUM.

A BRIEF GUIDE TO THE FOOD COLLECTION.

(FIRST ISSUE.)



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THE BETHNAL GREEN BRANCH OF THE SOUTH KENSINGTON MUSEUM.

UNDER THE DIRECTION OF THE LORDS OF THE COMMITTEE OF COUNCIL ON
EDUCATION.

Lord President, THE MARQUESS OF RIPON, K.G.

Vice-President, THE RIGHT HON. W. E. FORSTER, M.P.

I. In tracing the origin of the Branch Museum of Science and Art at Bethnal Green it will be necessary to refer, though briefly, to the early days of the parent institution, at South Kensington, from whence a considerable portion of the new edifice and of its contents have been derived.

II. The South Kensington Museum stands on 12 acres of land, acquired by the Government at a cost of 60,000*l.*, being a portion of the estate purchased by Her Majesty's Commissioners for the Exhibition of 1851, out of the surplus proceeds of that undertaking.

III. Here, in 1855, a spacious building was constructed, chiefly of iron and wood, under the superintendence of the late Sir William Cubitt, C.E., at a cost of 15,000*l.*, intended to receive several miscellaneous collections of a scientific character mainly acquired from the Exhibition of 1851, and which had been temporarily housed in various places.

IV. In addition to the collections already alluded to, the whole of the Fine Art collections which had been exhibited at Marlborough House since 1852 were also removed thither, and these were supplemented by numerous and valuable loans by Her Majesty the Queen and others.

V. This building was opened on June 22nd, 1857, as THE SOUTH KENSINGTON MUSEUM. Although in many respects well suited to its purpose, this iron building was avowedly of a temporary character, and from the first

it was intended to replace it by buildings of a more architectural character and of more substantial materials. The erection of these permanent buildings was commenced at once, and at the beginning of the year 1865 sufficient progress had been made to render the removal of the iron building necessary.

VI. It appeared to the Lords of the Committee of Council on Education that “this iron building might usefully be divided into three portions, and that one of these portions might be offered to the proper authorities in the north, east, and south of London respectively, at a nominal sum, in order to assist in the formation of district museums, security being required for the completion of each portion in a suitable manner, and for its permanent appropriation to public uses.” After some correspondence with other Departments of the Government, it was decided that measures should be taken for carrying out this proposal.

VII. On May 6, 1865, a meeting of noblemen and gentlemen interested in the establishment of Suburban or Metropolitan District Museums was held at the South Kensington Museum, the Lord President of the Council, Earl Granville, being in the chair, at which the proposal was fully discussed, and a strong desire was expressed by the representatives of the various suburban districts of the north, east, and south of London to secure a share of this building, the great difficulty felt in each case being the providing of a suitable site. It was decided that after a period of six months each district should be at liberty to put in its claim to a portion of the iron building, and send its application to the Science and Art Department.

VIII. On March 7th in the following year (1866) Mr. now Sir Antonio Brady addressed the following letter to the Lord President of the Council:

Stratford, E., 7 March 1866.

May it please your Lordship,

WHEN I and others acting with me had the honour of attending the meeting held under your Lordship’s presidency, in the Lecture Room of South Kensington Museum, on the 6th May last, on the subject of Local Metropolitan Museums, I put in a plea on behalf of the million artisans inhabiting the densely populated manufacturing and labouring

districts in the East of London; and I pointed to a site most admirably placed in the very centre of the East-end, which I then hoped might be made available for the proposed museum.

The land in question, containing about 4½ acres, is close to Mile-end Station, one mile from Shoreditch on the Great Eastern Railway; it is near the junction of the Hackney and Cambridge Heath Roads, and is the centre of a network of railways, and omnibuses run in all directions, at twopenny and threepenny fares to and from all parts of London.

The site is about one mile and three quarters from the Bank of England, and two miles from the General Post Office, and taking the proposed site as a centre, within a radius of two miles are comprised a large portion of the following extensive districts, viz.: City of London, Shoreditch, Finsbury, St. Luke's, Old Street, Hoxton, Islington, De Beauvoir Town, Canonbury, Ball's Pond, Kingsland, Dalston, Clapton, Homerton, Hackney, Victoria Park, Old Ford, Bow, Stepney, Limehouse, Poplar (including West India Docks), parts of Rotherhithe and Bermondsey (including Surrey and Commercial Docks), Shadwell, Wapping, St. George's-in-the-East (including London and St. Katharine's Docks), Tower, Whitechapel, and Mile-end.

This circle of two miles radius embraces the N.E. and E. postal districts, part of the N. district, and parts of the E.C. and S.E. districts.

The land in question was bought as a gift to the poor in King James's reign, when this part of London was open fields, and the trustees, with the consent of the Charity Commissioners, have unanimously agreed to sell the land for the purposes of the proposed museum, and to invest the purchase-money, which has been conditionally offered and accepted.

I have now the pleasure of informing your Lordship that, if this site is acceptable to the Government, I am authorised, on the part of the committee acting with me, to guarantee to raise the purchase-money necessary to acquire the fee simple, and to offer this magnificent site to the Government for the purpose of erecting thereon a museum for the East-end of London.

The site is marked red in the accompanying maps, and is more particularly described in the plan hereunto annexed; it will be seen that it occupies a most commanding position. There is no other suitable spot unbuilt on, but if we had the choice of any ground in the East-end we should recommend the position of this site in preference to any other.

It is not my purpose to enter on the advantages of local museums. After what passed at the meeting at South Kensington, the value of institutions such as we wish to see established in the East-end is admitted on all hands; but what I desire respectfully to submit to your Lordship is the kind of museum which those acting with me would wish to see erected.

During the past year the subject of local museums has been much ventilated, and as the time has now arrived when it seems a necessity to provide more room for the great national collections, we respectfully submit that it is a good and fitting opportunity to make the national collections more useful and more accessible than they now are, and I trust this splendid site may induce the Government to entertain the propositions I have the honour to submit for their consideration.

1st. From inquiries made since the meeting last May, it is found that it will be utterly and entirely impracticable for a permanent building to be erected by local efforts, or to maintain the necessary staff if a building were otherwise provided; and we feel that this could only efficiently be done by the Government as a part of one comprehensive scheme. We find it will require all our efforts to raise the funds to pay for the site, and under these circumstances we humbly submit to your Lordship that the Government should, in exchange for the site, take the whole matter into their own hands as a national affair.

2ndly. The scheme that commends itself most to our minds is, not to distribute the superfluities of the British Museum *piecemeal* amongst several local museums, but that typical collections illustrating one branch of science should be arranged in one of several museums in different quarters of the metropolis. The British Museum would thus

be relieved of its plethora without impairing the value of any one collection; for instance, the natural history collections may be kept together in one place, the ethnological in another, so that anyone requiring to study any particular branch would know to what museum to resort.

In any plan of a museum that may be adopted for the improvement of the working classes, we submit that if they are to benefit by it to the fullest extent, it must be placed in a neighbourhood accessible to them, and must be open of an evening. We submit that it be made *educational in the widest sense of the word*, and that convenient and comfortable refreshment-rooms be added to the other attractions of the place.

I am to request that your Lordship will be pleased to communicate your wishes in this matter, that we, on our part, may at once take the necessary means to give legal effect to this arrangement, if concurred in by your Lordship.

The land being unoccupied would be available immediately the preliminary agreements were finally settled.

I have, &c.

(Signed) Antonio Brady, J.P.,
Honorary Secretary.

To the Right Honourable Earl Granville, K.G., Lord President of Her Majesty's Most Honourable Privy Council.

IX. This letter was at once taken into favourable consideration by the Lords of the Committee of Council on Education, Earl Granville and Mr. Bruce being respectively President and Vice-President. A change of Government shortly afterwards took place which caused some delay, but on December 6, 1866, the Duke of Buckingham being President and Mr. Corry Vice-President, a minute was passed recommending the proposal to the favourable consideration of the Lords Commissioners of Her Majesty's Treasury, and asking that an estimate of the probable cost might be included

in the votes of the ensuing year. The following paragraph occurs in this minute:—

“My Lords regret that Mr. Brady’s offer on behalf of Bethnal Green can be adduced as the sole proof of the practical earnestness of the several districts of the metropolis to act in establishing district museums. Their Lordships, therefore, propose that the iron columns, flooring, stairs, window fittings, heating arrangements, &c. of the whole of the iron building should be re-erected as soon as practicable at Bethnal Green, on the free site provided by the locality, but that brick walls and a slate roof should be used instead of iron; and they estimate that the cost will be 20,000*l.* The works would thus be of a permanent nature.”

X. The Treasury (the Right Hon. B. Disraeli being then Chancellor of the Exchequer) accepted the proposal to re-erect the structure and to provide for its maintenance, and a vote of 5,000*l.* on account was granted by the House of Commons towards the cost of removal and re-erection of the building, but some delay arose in consequence of legal difficulties as to the conveyance of the ground. By the untiring efforts of Sir Antonio Brady, the Rev. Septimus Hansard, rector of Bethnal Green, Mr. J. M. Clabon, Dr. J. Millar, and others, heartily seconded by the trustees of the land and supported by the Government, these difficulties were at length surmounted, a special Act of Parliament having been obtained for the purpose (31 Vict. c. 8.), and on 13 February 1869, the four gentlemen above named, acting on behalf of the subscribers to the fund for the purchase of the site, attended at the Council Chamber, Downing Street, and presented to the Lord President and Vice-President of the Committee of Council on Education the title-deeds of the site. ^[7]

XI. After the removal of the materials had taken place the erection of the building was at once commenced in accordance with plans prepared for the Department of Science and Art under the direction of Major-General Scott, C.B.

XII. At the beginning of the present year (1872) the building was sufficiently advanced for the reception of objects. Two important collections, formerly exhibited in the iron buildings, already existed in the

South Kensington Museum ready for transfer to Bethnal Green, the ANIMAL PRODUCTS COLLECTION intended to illustrate the various applications of animal substances to industrial purposes, and the FOOD COLLECTION, one of the most popular divisions of the Museum. These, with an important series of examples of Economic Entomology recently formed by Mr. Andrew Murray, now occupy the whole of the space on the ground floor under the galleries, and it is confidently believed that they will prove of great and abiding interest and educational value, forming as they do no inconsiderable contribution towards the establishment of a complete trade museum, the necessity for which at the East-end of the metropolis has long been recognised.

XIII. The galleries of the building on the first floor are at present assigned to Paintings and other Fine Art objects, and the Lords of the Committee of Council on Education are indebted to the generous liberality of Sir Richard Wallace, Bart., for the loan of a collection of Art Treasures of almost unexampled beauty and value, occupying the whole of the space assigned to this division. These Art Treasures, collected by the late Marquis of Hertford, K.G., during a period of 30 years, have hitherto been comparatively unknown to the English public, a large portion of the objects having been specially brought over from Paris within the last three months at the expense of Sir Richard Wallace.

XIV. The basement of the building contains a range of spacious and well-lighted rooms. A portion of this will serve as Refreshment Rooms, and it is proposed to use the remainder for educational purposes, including a Library, and rooms in which classes may receive instruction in the various branches of Science and Art.

XV. It was desired by Her Majesty the Queen that on the 24th June 1872 the Museum should be opened in state by His Royal Highness the Prince of Wales on behalf of Her Majesty, the Prince being accompanied by Her Royal Highness the Princess of Wales.

HENRY COLE,
Director.

The **Bethnal Green Branch** of the **South Kensington Museum** was opened to the public on **Tuesday**, the **25th June 1872**, under the following regulations, which are the same as those of the **South Kensington Museum**:—Daily (except **Sundays**). **Free admission** on **Monday**, **Tuesday**, and **Saturday**, from 10 a.m. to 10 p.m. On **Wednesday**, **Thursday**, and **Friday (Students' days)**, admission **sixpence**, from 10 a.m. to 4, 5, or 6 p.m., according to the season.

Tickets of Admission on Students' days (available both for the Bethnal Green Museum and the South Kensington Museum) are issued at the following rates:—*weekly*, 6d.; *monthly*, 1s. 6d.; *quarterly*, 3s.; *half-yearly*, 6s.; *yearly*, 10s. *Yearly* Tickets are also issued to any school at 1l., which will admit all the pupils of such schools on all Students' days. To be obtained at the Catalogue Sale Stall of each Museum.

JULY 1872.

A BRIEF GUIDE TO THE FOOD COLLECTION. [\[9\]](#)

** * An Inventory of the Collection, arranged alphabetically, and containing fuller information than this Guide, can be obtained at the Catalogue Stall, Price Sixpence.*

THE idea of the Food Collection (originally formed in 1857), now removed from the South Kensington Museum and arranged in the lower Gallery on the North side of the Branch Museum at Bethnal Green was suggested by Thomas Twining, Esq., of Twickenham, as part of a plan for the establishment of an Economic Museum that should comprise illustrations of every-day life for the working classes. The Food Collection was at the commencement of its formation carried on under the direction of Dr. Lyon Playfair, M.P., and, as now constituted, has been arranged with the express object of teaching the nature and sources of the food which rich and poor alike need for the maintenance of their daily life. Considerable progress has been made in carrying out this design, and the present brief Guide is intended as an introduction to the general principles and plan upon which the Collection has been arranged. Two great objects have been kept in view in the Collection:—

First, to represent the chemical compositions of the various substances used as food; and, secondly, to illustrate the natural sources from which the various kinds of food have been obtained. Where the processes of the preparation of food admit of illustration, these are also exhibited.

There are many methods by which such a Collection might be arranged; but the Chemical Composition of Food has recently been discovered to have so close a connexion with its action on the system, that it has been deemed

advisable to follow a *Chemical arrangement*. All food is found to be composed of the same materials or elements as the Human Body. The necessity of the supply of food from day to day depends on the fact, that the elements of the human body are daily wasted by the processes of life. As a fire cannot *burn* without a supply of *fuel*, neither can the human body *live* without its daily supplies of *food*.

COMPOSITION OF THE HUMAN BODY.

Not only does food supply the daily waste of the human body, but, as the body increases in size from birth to adult age, it is supplied with materials for this increase by the aid of food. In order, therefore, to understand the value of food from its composition, it is necessary to know the composition of the human body. Just as any other compound substance can be submitted to chemical analysis and the elements of which it consists ascertained, so can the composition of the human body be discovered. Such analyses of course become difficult in proportion to the complication of the body analysed, and only an approach to the true quantities in which the elements exist can be expected. In Case No. 1, Division A., the results of such an analysis have been attempted, and the quantities of each element entering into the composition of a human body weighing 11 stone or 154 pounds are (as far as possible) presented to the eye.

The following are the elements and their quantities:—

ULTIMATE ELEMENTS OF THE HUMAN BODY.

	lbs.	ozs.	grs.
1. <i>Oxygen</i> , a gas. The quantity contained in the body would occupy a space equal to 1,314 cubic feet	111	0	0
2. <i>Carbon</i> , a solid. When obtained from animals it is called animal charcoal	21	0	0
3. <i>Hydrogen</i> , a gas. The lightest body in nature. The quantity present would occupy about 2,622 cubic feet	14	0	0

4. <i>Nitrogen</i> , a gas. It would occupy, when free, about 46 cubic feet	3	9	0
5. <i>Calcium</i> , a solid. The metallic base of lime which has not yet been obtained in sufficient quantity to be employed in the arts. It is about the density of aluminium	2	0	0
6. <i>Phosphorus</i> , a solid. This substance is so inflammable that it can only be kept in water	1	12	190
7. <i>Chlorine</i> , a gas. When combined with sodium it forms common salt	0	2	382
8. <i>Sulphur</i> , a solid. A well-known substance. It unites with hydrogen, forming sulphuretted hydrogen, which gives the unpleasant smell to decomposing animal and vegetable matter	0	2	219
9. <i>Sodium</i> , a metal. It is so light that it floats on water, and is kept in naphtha to prevent its oxidation	0	2	116
10. <i>Fluorine</i> , a gas. This substance has not been separated in such a manner as to permit of an examination of its properties, and cannot be exhibited. It is found united with calcium in the bones	0	2	0
11. <i>Potassium</i> , a metal. Like sodium it floats on water, and burns with a flame when placed on it	0	0	290
12. <i>Iron</i> , a metal. In small quantities it is necessary to the health of the body	0	0	100
13. <i>Magnesium</i> , a metal. Combined with oxygen it forms magnesia	0	0	12
14. <i>Silicon</i> , a non-metallic substance. With oxygen it forms silex or silica. It enters into the composition of the teeth and hair	0	0	2
	154	0	0

Other elements have been found in the body, as copper and manganese, but these are probably accidental.

These elements, when combined together, form a set of compound bodies called “proximate principles,” out of which the tissues and fluids of the body are formed.

PROXIMATE PRINCIPLES OF THE HUMAN BODY.

	lbs.	ozs.	grs.
1. <i>Water</i> , composed of oxygen and hydrogen gases	111	0	0
2. <i>Gelatin</i> , of which the walls of the cells and many tissues of the body, as the skin and bones, are principally composed	15	6	0
3. <i>Fat</i> , which constitutes the adipose tissue	12	0	0
4. <i>Phosphate of Lime</i> , forming the principal part of the earthy matter of the bones	5	13	0
5. <i>Fibrin</i> , forming the muscles and the clot and globules of the blood	4	4	0
6. <i>Albumen</i> , found in the blood and nerves	4	3	0
7. <i>Carbonate of Lime</i> , also entering into the composition of bone	1	0	0
8. <i>Chloride of Sodium</i> , common salt	0	3	376
9. <i>Fluoride of Calcium</i> , found in the bones	0	3	0
10. <i>Sulphate of Soda</i>	0	1	170
11. <i>Carbonate of Soda</i>	0	1	72
12. <i>Phosphate of Soda</i>	0	0	400
13. <i>Sulphate of Potash</i>	0	0	400
14. <i>Peroxide of Iron</i>	0	0	150
15. <i>Phosphate of Potash</i>	0	0	100
16. <i>Phosphate of Magnesia</i>	0	0	75

17. <i>Chloride of Potassium</i>	0	0	10
18. <i>Silica</i>	0	0	3
	154	0	0

These compounds, in passing away from the body, form many others, which may be here left out of consideration as not forming a necessary part of the fabric of the human body.

None of these constituents of the body remain permanently in the system, and whilst the old particles are being removed new ones are supplied by the food. It is calculated that in this way a quantity of material, equal to the weight of the whole body, is carried away every forty days. So that we may be said to moult or cast away our old body and get a new one every forty days.

The materials for the food of man, and containing the above elements, are derived from the mineral, vegetable, and animal kingdoms. The vegetable kingdom, however, is the great source of food to man and animals, as it is in the cells of the plant that the elements undergo those chemical changes which fit them for food. The animal can only supply what it obtains from them, and the substances supplied by the animal kingdom as food are identical with those obtained from plants.

The FOOD COLLECTION is arranged as nearly as possible in the order of the following CLASSIFICATION, commencing at the *western end* of the lower gallery.

CLASS I. ALIMENTARY OR NECESSARY.

Group 1. *Mineral Substances.*

Examples: Water; Common Salt; Ashes of Plants and Animals.

Group 2. *Non-nitrogenous force-producing Substances, incapable of forming Flesh or Muscle.*

Examples: Sago, Arrowroot (*Amylaceous*); Sugar, Figs, Dates (*Saccharine*); Animal and Vegetable Fats and Oils (*Oleaginous*).

Group 3. *Nitrogenous Substances capable of producing both Flesh and Muscle.*

Examples: Eggs (*Albuminous*); Wheat, Flesh (*Fibrinous*); Peas, Cheese (*Caseinous*).

CLASS II. MEDICINAL OR AUXILIARY.

Group 1. *Containing Alcohol.*

Examples: Beers, Wines; Spirits.

Group 2. *Containing Volatile Oils.*

Examples: Spices and Condiments, as Cloves, Nutmegs, Pepper, Horse-radish, &c.

Group 3. *Containing Acids.*

Examples: Apples, Oranges, Rhubarb Stalks, Vinegar.

Group 4. *Containing Alkaloids, which act upon the nervous system as stimulants or sedatives.*

Examples: Tea, Coffee, Cocoa; Tobacco, Hemp, Opium.

CLASS I.—ALIMENTARY OR NECESSARY. WATER.

The first and most essential constituent of food is water. Three fourths of the body is composed of water, and it is by the agency of water that all kinds of food are taken up into the system. Solid food contains large proportions of water, but, in proportion to the dryness of food, water should be added to it, in the form of some kind of beverage.

QUANTITIES OF WATER IN 100 POUNDS OF DIFFERENT KINDS OF SOLID FOOD.			
<i>Vegetable Food.</i>			
	lbs.		lbs.
Cabbage	92	Maize	14

Turnips	87	Peas	14
Carrots	86	Beans	14
Beetroot	83	Lentils	14
Parsnips	79	Buckwheat	14
Potatoes	75	Oatmeal	13
Bread	44	Rye	13
Flour	14	Rice	13
Barley Meal	14	Cocoa	5
<i>Animal Food.</i>			
Milk	86	Lamb	50
Eggs	80	Mutton	44
Fish	78	Cheese	40
Veal	62	Pork	38
Beef	50	Bacon	30

An imperial gallon of water weighs 10 pounds avoirdupois weight.

Water for dietetical purposes is obtained principally from three sources:—1. Rivers; 2. Surface wells; 3. Deep or Artesian wells. Water from all three sources contains saline or mineral matters in solution, and, provided they are not in quantities so large as to act injuriously on the system, water may become a source of supply of these constituents to the body. The best remedy for impure water is filtering, which may be done by passing the water through charcoal and sand. “A Poor Man’s Filter” is exhibited in the Museum, which can be very easily and cheaply constructed by using a common flower-pot, glazed inside, plugging the drainage hole (not too tightly) with a piece of clean sponge, then adding layers of animal charcoal, sand, and rather coarse gravel. Filters from the establishments of the Messrs. Lipscombe, the Messrs. Ransome, and the Carbon Filter Company are also exhibited. The passing water over iron has been found to have a remarkably purifying effect, and this has been patented by Dr. Medlock.

The *organic* impurities of water are best tested by the aid of the microscope, but, as an examination by this instrument requires much time, a ready method of obtaining a knowledge of the comparative organic impurity of waters is the addition of the permanganate of soda or potash. This salt, which gives to water a beautiful red colour, is easily decomposed by organic matters. When the same quantity of the permanganate is added to a series of waters containing organic matters, those which contain the least retain the most colour and *vice versâ*. Waters thus tested are exhibited in the Collection.

Water from the chalk or limestone is generally hard, arising from its holding in solution carbonate of lime, which, although insoluble in water, is dissolved by the agency of carbonic acid. By Clark's softening process the carbonic acid is neutralized by lime, and the carbonate of lime is thus thrown down. Specimens are exhibited.

Water is frequently stored in leaden cisterns, and when free from carbonic and phosphoric acids it acts powerfully on lead. Thus distilled water becomes speedily tainted with lead, whilst Thames water and London surface well water act but slightly upon it. Specimens of these waters acting upon lead are exhibited in the Collection.

SALT.

Common Salt is a chloride of sodium, and exerts an extraordinary influence on animal as well as vegetable life. All marine animals and plants seem to have their existence determined by this substance. It enters into the composition of the human body, and all over the world man uses it, when he can obtain it, in its mineral form, as an addition to his food.

In Case 5 will be seen a collection of salt from various countries.

MINERALS IN FOOD.

In Case 4 are examples of some of the principal Mineral Substances, excepting water, in food. They are generally essential to proper nutrition. In the body of a man, weighing 154 lbs., there are about 8 lbs. of mineral matter. Different parts of the body show peculiar affection for particular

ingredients to the exclusion of others. The mineral salts contained in plants and animals are indestructible by heat, hence they are called “ashes.”

It should be recollected, that in the boiling of food many of the mineral substances are dissolved out of it, and where the liquid that they are boiled in is not consumed such mineral matters are thrown away. This is the case with boiled meat and vegetables, and a constant use of such food may lead to injurious effects. The best corrective to such a diet is the use of uncooked fruit and vegetables. In this way the eating of ripe fruits, as apples, pears, gooseberries, &c., and salads, has a beneficial effect on the system.

SALADS.

Although many things eaten as salads contain other constituents of food besides mineral matters their beneficial action in diet is due to the latter. The practice of eating salads is not so common in Europe as before the introduction of the potato, which, to a certain extent, supplies the same kind of mineral matters to the blood. The practice of eating salads is, nevertheless, to be highly commended; and many plants formerly much used might now be consumed in this way with much advantage. Plants thus used contain a larger quantity of mineral matter than vegetables which have been boiled in water. Various plants used as salads may be seen on the shelves, and are renewed from time to time.

FORCE-PRODUCERS IN FOOD.

In Case 4 may be seen the varieties of substances called “force-producers in food.”

STARCH.

The substance called Starch is found very abundantly in the vegetable kingdom. Its presence was at one time regarded as characteristic of plants, but it has recently been found in animals. It occurs in the form of irregularly-shaped granules, which vary in size from the $\frac{1}{400}$ to the $\frac{1}{2000}$ -th of an inch in diameter. These granules are simple or compound. They vary in shape and size in every species of plant, and are insoluble in

water, but are easily diffused through it. On being mixed with water, and exposed to a temperature of 180°, the starch gelatinises, and, mixing with the water, thickens it. This occurs in the cooking of starch, and this property lies at the foundation of pudding making.

Starch is abundantly present in all the more common forms of vegetable diet. It exists in a state of almost absolute purity in the substances known as arrowroot, tapioca, and sago. These substances from whatever source obtained, contain little or no nutritious or flesh-forming food, and, consequently, ought never to become the substantive diet of human beings. Many plants contain so large quantities of starch, and so small quantities of flesh-forming matter, that they ought only to be taken on account of their starch. Such are the potato and rice, in which the quantity of starch to flesh-forming matter is as 14 to 1, whilst in wheat it is only as 5 to 1. Potatoes and rice, therefore, can never form the staple article of the diet of the people of this country, who need a large quantity of force-producing matter in order to enable them to perform their work.

Starch is extensively used in the arts manufactures, and for domestic purposes. It is prepared for this purpose from the potato, wheat, rice, flour, and the coarser kinds of sago.

In Cases **6** and **16** is an extensive series of starches, sago, arrowroot, tapioca, &c. &c., from various parts of the world.

The following table gives the quantities of Starch in 100 parts of various kinds of food:—

Rice	74	Beans	36
Maize	60	Lentils	35
Wheat	59	Parsnips	17
Rye	51	Potatoes	15
Buckwheat	50	Mangel Wurzel	12
Bread	48	Carrots	11
Barley	48	Turnips	10

Oats	39	Cabbage	4
Peas	37		

SEA-WEEDS USED AS FOOD.

Sea-weeds contain lichen starch, and are frequently used as food. Specimens may be seen in Case 7. In China the people are very fond of sea-weeds, and many kinds are collected and added to soups, or eaten alone with sauce. In times of scarcity the poorer inhabitants of the sea-shores of Europe have recourse to sea-weeds for a supply of food.

THE POTATO.

Although this plant contains but a small quantity of flesh and force-producing matter, it yields an abundance of starch and mineral matters in a condition which acts very beneficially on the human system, and its introduction into Europe has been of the greatest benefit to its teeming populations.

The potato is an herbaceous plant producing annual stems from an underground tuber or root-stock which is the part that is used as an article of food. It has white flowers and a green fruit, which, like all the plants of the order to which it belongs, contain a poisonous principle. The native country of the plant is South America. It has been found wild in various parts of Chili, and also near Monte Video, Lima, Quito, Santa Fe de Bogota, and in Mexico. Spain was the country in which this plant was first cultivated in Europe; from thence it extended into Italy. It was first grown in the British Islands by Sir Walter Raleigh in his garden at Youghal in Ireland, but it was not generally cultivated in Great Britain till the middle the last century. The only part of the plant employed as food is the tuber, which is a kind of underground stem. Upon this stem buds are formed which are called "eyes," and from these, by cutting up the potato, the plant is propagated. The tubers of the wild potato are small in size, but by culture they may be very much enlarged. In this country many varieties of the potato are known under the names of "kidneys," "rounds," "reds," "blues," "whites," &c. Many of these varieties are now disappearing, the "white," "kidney," and "round" potatoes being preferred to all others. The potato contains large

quantities of water (75 per cent.), and less flesh and force-producing matters than any other plant cultivated for human food. It is therefore not adapted for consumption as a principal article of diet, and should only be employed as an addition to more nutritious kinds of food. It contains a variety of mineral matters, which also render it valuable as an article of diet. It has for many years been liable, in Europe, to a diseased condition, in which the water seems to be increased, and decomposition consequently readily sets in. The decayed parts are infested by a fungus, but this has not been shown to have anything to do with the production of the disease. Potatoes are largely employed in this country for the production of starch, which is used for a variety of purposes in the arts and manufactures. Potatoes are cooked in many ways, and all the varieties of food which can be obtained from the flour of the cerealia may be procured from the potato, as starch, macaroni, vermicelli, &c.

The analysis of the Potato may be seen in Case **8**, as well as various preparations from it.

RICE.

This plant belongs to the natural order of grasses. It is a native of East India, and is extensively cultivated throughout Asia, in North and South Carolina, and other parts of the world. Although more largely consumed by the inhabitants of the world than any other grain, it contains less flesh and force-producing matter.

When employed in this country it should only be used as an adjunct to other kinds of food more rich in force-producers. Boiled, as an addition to meat, or in the form of pudding or curry, it may be judiciously employed, as a variety, especially in the food of the young.

Case **9** shows the analysis of rice, and many samples of the grain are exhibited in the Collection.

SUGAR.

Sugar has a chemical composition very nearly resembling starch, but it differs in both chemical and physical properties. Sugar is soluble in water, whilst starch is only diffusible through it. Sugar undergoes the process of

fermentation, which starch does not. Sugar has a sweet taste, while starch is almost tasteless. Starch is, however, convertible into sugar by the agency of nitrogenous substances. If starch is placed in contact with saliva a little time it becomes soluble, and gives the reactions of sugar; and it is probable that in this way starch itself becomes absorbed into the blood. Sugar, like starch, assumes various forms, and three of these are found in common articles of diet. These are cane sugar, grape sugar, and milk sugar.

The action of sugar on the system is identical with starch. As it is more readily absorbed into the blood than starch, it is better adapted as a force-producer for the young. Hence it is found supplied to the young in all the mammalia, in the milk secreted by their mothers. That it is adapted for the young is shown by the instinctive propensity children display to partake of this form of diet. Although adapted for children, the facility with which it decomposes renders it frequently injurious to adults.

Most plants contain sugar in their roots. But in some large quantities are deposited, as in the sugar beet, which is employed most extensively in France and on the continent of Europe for the supply of sugar for dietetical purposes. A series of specimens illustrating products from beet-root, including sugar, from Messieurs Serret, Hamoir, and Co., of Valenciennes, are exhibited in the Collection.

Sugar is the basis of all kinds of confectionery, specimens of which are exhibited by Messrs. Fortnum and Mason, of Piccadilly, in Case **111**.

Samples of sugar are exhibited in Cases **17**, **110**, **112**, and **113**, some of which have obtained been from other plants than the sugar cane.

Fruits after being saturated with sugar are also preserved and kept dry. In Case **15** preserved fruits of various kinds are exhibited by Messrs. Fortnum and Mason. It is in this way that fruits are brought to this country which otherwise would not be seen on account of their perishing nature.

Treacle or *Molasses* is the uncrystallized portion of sugar which is separated by draining from the brown sugar.

Grape Sugar or *Glucose* is found in the fruits of plants, and is especially abundant in the grape. Grapes, when dried, are eaten on account of the glucose they contain. They are known in the shops under the name of

“plums,” “raisins,” and “currants.” The latter word is a corruption of Corinth, the small grape yielding this, being cultivated in the vicinity of Corinth, on the classic soil of Greece.

Dried fruits of the grape-vine, presented by Messrs. Fortnum and Mason, Piccadilly, are exhibited in Case **14**.

Honey, which is the stored food of the bee, contains both crystallizable and uncrystallizable grape sugar. The crystals of the former may be easily detected by the aid of a low power of the microscope. Samples of British honey, and honey from France, Russia, and other countries, may be seen in Cases **18**.

Substances resembling Sugar, such as dextrin, gum, liquorice, manna, &c., are exhibited in Case **107**. Among plants yielding sugar may be noted the Chinese sugar millet (Case **17**), sweet potato (Case **8**), turnips (Case **11**), carrots (Case **11**), and Jerusalem artichoke (Case **12**), the analyses of which are exhibited.

FAT AND OIL.

Under the names of oil, butter, fat, lard, suet, and grease, a substance is used largely as an article of food, which differs from starch and sugar in the absence of oxygen gas. The composition of these oleaginous substances may be represented as follows:—Carbon 11 parts; hydrogen 10 parts; oxygen 1 part.

Oil differs from the other carbonaceous substances in food in not only supplying materials for maintaining animal heat, but in forming a part of the tissues of the body called fat. The quantity consumed in animal food is very large, constituting frequently more than half of the bulk of the food consumed. It is also found very generally present in the vegetable substances used as food. Although essential as an article of diet in certain quantities, oil is less digestible than other kinds of food, and those foods which contain it in large quantities are generally indigestible. The principal source of oil used as food from the vegetable kingdom is the Olive. This plant is cultivated in the south of Europe. The part of the plant which contains the oil is the fruit. The seeds of most plants contain oil in addition to starch and other principles. Many seeds are used for obtaining oil for

various purposes in the arts, as the poppy, rape, mustard, hemp, and flax seeds. In Case **20** is a collection of nuts and seeds containing oil commonly eaten as food. Case **21** contains the analysis of the coco-nut; and in the same Case that of an African bread called “Dika bread,” both of which illustrate food products containing an abundance of oil. The cocoa, or chocolate plant, is one of the most remarkable vegetable productions yielding oil, the seeds giving nearly 50 per cent. of a hard oil, or butter. See Case **53**.

FLESH AND FORCE-PRODUCERS.

In Case **4** are shown those ingredients of food, which are capable of forming muscle or flesh. They are made use of in the human body partly for the construction of muscle, and partly for the production of mechanical force and heat. They are all nearly identical in their chemical composition.

1. ALBUMEN, made from Eggs and from Blood. It forms about 7 parts in 100 of blood, and is always present in lymph and chyle. Liquid or soluble albumen, as shown in the white of egg, coagulates by heat and various chemical agents.
2. ALBUMEN, as found in the juices of carrots, turnips, and cabbages, and obtained by boiling their juices. It is the same body as albumen from eggs.
3. FIBRIN made by stirring blood with a rod. It is the basis of muscle or flesh. Flesh-fibrin probably bears the same relation to blood-fibrin as coagulated albumen does to soluble albumen.
4. FIBRIN made from Wheat-flour. It is identical with the fibrin found in flesh, but not exactly the same as that found in blood, and is known as *Gluten*.
5. CASEIN prepared from milk, in which it is soluble, owing probably to a little alkali: when an acid is added, the Casein curdles or coagulates, and then is known as Cheese. In 100 parts of cows' milk there are 3½ parts of Casein.
6. CASEIN or LEGUMIN as found in peas, beans, lentils, coffee, &c. The Casein of Vegetables is now supposed by most chemists to be identical with

the Casein or Cheese of Milk, but a few chemists still deny this. 100 parts of peas contain above 20 parts of Casein.

EGGS.

Eggs are very nutritious articles of food. They contain as much oil or fat and flesh and force-producing matter as butcher's meat. The white is not, however, so digestible as the flesh of meat. They enter into the composition of puddings, cakes, buns, and other forms of diet. They are also eaten alone, boiled or fried, and are most digestible when least done.

The egg of the domestic fowl is usually eaten, but those of other birds are frequently employed as food. The eggs of the woodcock, plover, and other small birds, are esteemed a luxury. Those of the duck and goose have a strong flavour, and those of sea-fowl are fishy. The eggs of the turkey are rich in flavour, whilst those of the guinea-hen have a very delicate flavour. All birds' eggs may be eaten with impunity. The eggs of the crocodile, and other oviparous reptiles, are eaten in some parts of the world.

In Case **61** is a collection of the Eggs of domestic poultry and some other birds, together with the analysis of Hen's Eggs.

The Flesh and Force-producers are most abundant in those plants which yield the substantive food of man. These plants belong principally to the group of cereal grasses, as wheat, oats, barley, &c., and leguminous plants, as peas, beans, lentils, &c. Of these the most important is wheat. At the western end of the gallery are shown samples of many cultivated varieties of wheat, oats, barley, rye, and maize, in the straw, and in grain.

The Flesh and Force-producers exist also in large quantities in milk, and in the flesh of vertebrate animals, divided into mammals, birds, fishes, and reptiles.

WHEAT.

The wheat plant is grown all over the world, but flourishes mostly between the parallels of 25 and 60 degrees of latitude. It is more abundant in the northern than in the southern hemisphere.

The varieties of wheat cultivated in Europe may be divided into those whose flowers produce awns, and those without these appendages, or *bearded* and *beardless* wheats. The fruits or seeds of these varieties are red or white, hence a further subdivision takes place into *red* or *white*, bearded or beardless, wheats. Amongst the red bearded varieties is the fingered Egyptian or Mummy Wheat, which presents the peculiarity of several branches bearing fruits proceeding from its central stalk. Wheat is also called hard and soft according to its consistence, and winter and spring as it is sown at those seasons of the year. The red varieties yield the largest amount of grain, but the white the whitest flour.

Wheat is preferred to the other cereal grasses as an article of food on account of its containing a larger quantity of flesh-forming matters. The flour also may be rendered very white by separating it from the husks, or bran, and the fruit is much more easily separated from the chaff than is the case with the other cereals. The proportion of flesh and force-producing to those of force-producers only, is more nearly adjusted to the requirements of the system in wheat than in any other food. Hence, probably, its very general use as an article of food amongst the populations of the hardest working nations in the world.

In Case **26** is an analysis of the various constituents found in a pound of wheaten flour.

The chemical analysis of barley, buckwheat, maize, millet, oats, rye, and rice may be inspected in the respective cases.

BREAD.

The most common as well as the most important form in which wheaten flour is consumed as food is bread. In Case **25**, which may be called the "Bread Case," the constituent ingredients, with their respective quantities used in making bread, are exhibited. There are three methods of making bread, the ordinary or fermented process, the unfermented process, and that employed in making aërated bread. Bread is either *vesiculated* or *unvesiculated*, the latter is called unleavened bread, and consists of bread, and of such preparations of flour as are known by the names of biscuits, cakes, &c. of which two cases of samples are shown by Messrs. Peek,

Frean, and Co. of London, and J. W. Mackie and Sons of Edinburgh. For other details concerning bread the visitor is referred to the printed labels in the case.

ANIMAL FOOD OR FLESH.

According to the classification of the Food Collection, Flesh is placed next to Wheat and other cereals in Group 3, which includes *nitrogenous substances capable of producing both flesh and force*.

Animal food is composed of the same materials as vegetable food. It is formed of the same elements, and presents the same proximate principles. It contains water and mineral matters of the same kind as plants. Its force-producing substances appear in the form of fat, and its flesh and force-producing substances in the form of fibrin and albumen.

MILK.

Of all animal foods milk is the most important, as it may be regarded as the type of human food. Case 55 contains an analysis of cow's milk, human milk, and asses milk, and is accompanied with explanatory labels.

Milk is preserved in various ways, so that it may be taken on long voyages or otherwise employed as a diet where living animals cannot be kept to produce it. It is preserved both in a liquid and solid state. The latter mode of preparation appears to have the advantage.

Butter is formed from cream by the process of "churning." The casein is held in solution in the milk by the aid of certain salts; when these are removed by acids the casein coagulates, and forms "curds." When the curd is removed with the butter and pressed it forms cheese. The best and highest-priced cheeses are those in which there is most butter. The casein without the butter is hard and indigestible.

THE FLESH OF ANIMALS.

At the western end of the gallery over the upright cases containing wheat, barley, oats, maize, &c., are arranged some selected heads of oxen in illustration of the principal breeds in this country.

The Case, **56**, is specially devoted to the composition of one pound of beef, mutton, pork, veal, lamb, and fowl. Wax models represent the substances, and each analysis is accompanied with descriptive printed labels.

In Case **70** are seen mounted specimens of the varieties of hares and rabbits indigenous to the United Kingdom.

The flesh of birds, fish, and reptiles is also represented in the Collection.

Cases **63** and **64** contain mounted examples of the varieties of pheasant successfully introduced into Great Britain, and a series of grouse, ptarmigan, capercailzie, &c., as representing the game birds of that family.

FISH is represented in the Collection by mounted specimens of the commoner kinds of fish brought to market, and by the analysis of a pound of salmon, mackerel, sole, conger eel, herring, and pike.

FISH yield a larger number of species used as food by man than either birds or quadrupeds. There are but few fishes caught in the fresh waters and seas of Great Britain that may not be eaten with impunity. In some countries the only animal food known is fish. The flesh of fish contains less oil or fat, and a larger quantity of mineral matters than the flesh of birds or mammals. The digestibility of fish is not so great as that of butcher's meat; hence, generally, it is not so nutritious as the flesh of birds or quadrupeds. Fish is undoubtedly a valuable as well as an agreeable article of diet, and should, where possible, be introduced into all dietaries.

In connexion with fish the collection illustrating Economic Fish Culture, mainly belonging to, and superintended by, Mr. Frank Buckland, Her Majesty's Inspector of Salmon Fisheries, should not be left unmentioned, although not forming a part of the Food Collection in the Branch Museum at Bethnal Green. This collection illustrates the science of breeding salmon, trout, and other fish by artificial means. It also includes a large number of casts of different kinds of fish, and a series of nets and other apparatus used in the legal and illegal capture of fish. At present it is exhibited in the arcades on the western side of the Royal Horticultural Gardens at South Kensington. During the period of the International Exhibition it is not accessible except to visitors to the Exhibition; but when the Exhibition is not going on, visitors can see the Museum of Economic Fish Culture under

the rules and regulations which govern the South Kensington Museum, with which it is officially connected as an addition to the Food Collection.

Lobsters, crabs, prawns, and shrimps, are exhibited in the collection in illustration of the edible animals belonging to the crustacea; and of molluscous animals, embracing the shell fish of the rivers and oceans, examples of the oyster, scallop, whelk, periwinkle, common snail, and Roman snail, are shown. (See Cases **59** and **60**.)

Of reptiles, but few are eaten in this country as food. Their flesh is, however, white and delicate, and rich in gelatin and fat. No accurate analysis seems to have been published of the flesh of these animals. The flesh of the green turtle is consumed in considerable quantity, and of it the famous turtle soup is made. The common and edible frog are eaten on the continent; and the land tortoise, common on the coast of the Mediterranean, is eaten by the inhabitants of Italy and the Levant. The flesh of the crocodile, alligator, and iguana, is also consumed in the countries where such creatures abound.

PEAS, BEANS, AND CHEESE.

These substances are next in the order of classification by reason of the large quantity of a flesh and force-producing substance contained in them called *casein*. The casein of vegetables is now supposed by most chemists to be identical with the casein or cheese of milk. The constituents or ingredients in one pound of peas, and in one pound of beans, are shown in Cases **30** and **31**. The visitor may examine a collection of beans from various foreign countries arranged in the Cases at the western end of the Collection.

Lentils are shown in Case **30**. By examining the analysis of this extremely nutritious product, the large proportion of casein is at once perceived.

CLASS II.—MEDICINAL OR AUXILIARY.

The Food Substances in this Class are divided into *four* groups, namely:—Those containing alcohol, those containing volatile oils, those containing

acids, and those containing alkaloids, which act upon the nervous system as stimulants or sedatives.

BEER.

The most common form in which alcohol is employed in this country is that of beer. Beer is distinguished from other alcoholic beverages by the addition of hops, which are the female catkins of a plant extensively grown in this country. Case **35** contains the materials from which porter is brewed, and illustrates the changes which malt undergoes during its conversion into beer. The analysis of stout, porter, pale ale, mild ale, and strong ale are exhibited in Case **35**, and the quantities of water, alcohol, sugar, and acetic acid are shown in an imperial pint of each. Next to beer the beverages containing alcohol consumed in this country are wines. The visitor is referred to the printed descriptive labels for details concerning the sources and manufacture of various wines from the juice of fruits, especially grapes. In Case **36** the quantities of water, alcohol, sugar, and tartaric acid, are shown in an imperial pint of the following wines:—Port, Brown Sherry, Pale Sherry, Claret, Burgundy, Hock, Moselle, Champagne, and Madeira.

In this country wines are made from oranges, raisins, gooseberries, currants, elderberries, and other fruits. They are usually called “home-made” or “British” wines. They contain other acids besides tartaric, hence the necessity of adding to them large quantities of sugar to cover the taste of the acid. Samples of British wines are exhibited in the Collection.

DISTILLED SPIRITS.

The terms “distilled” and “ardent” spirits are applied to alcoholic beverages which contain a very large per-centage of alcohol.

Those most commonly used are Gin, Rum, Whisky, and Brandy.

Alcoholic drinks when taken into the stomach act injuriously upon the mucous membrane, and when absorbed into the blood excite the nervous system. When taken in too large quantities, or upon an empty stomach, they lay the foundation of diseases of the stomach and surrounding organs, which often terminate in death. Their action on the nervous system, though

pleasant and agreeable, and even healthful in small quantities, becomes a source of fearful disease when carried to excess.

It is difficult to procure alcohol pure, and distilled spirits always contain a certain quantity of water. A spirit having a density of .920 is called “proof spirit” in this country; and when distilled spirits contain more or less alcohol than this, they are said to be “*under*” or “*above*” proof.

The quantities of alcohol, water, and sugar in an imperial pint of Brandy, Rum, and Gin, are shown in the Case **39**.

The physical degeneracy and moral degradation attendant upon taking alcohol in excess are well known; and no language is too strong to condemn the folly and wickedness of those who thus convert one of the blessings of Providence into a curse.

In Cases **38** the visitor will see samples of alcohol obtained from various sources; also a variety of flavored spirits, or “liqueurs,” from various countries, and a Japanese spirit or liqueur, distilled from rice, called “soke,” or “saki.”

VOLATILE OILS.

There is a large class of substances which are added to food for the purpose of giving it flavour, and which on account of the volatile oils they contain act as stimulants. These substances are known as Spices and Condiments. They also serve as the basis of a large number of sauces, which are sold ready prepared for the purpose of being added to cooked food. There is some difficulty in separating Spices from Condiments, but the former are more generally eaten with sugar, the latter with salt.

In Cases **40** to **43** will be found an extensive series of Spices and Condiments from various parts of the world.

FLAVORERS.

There is another class of substances, which cannot be called either condiments or spices, but which are extensively employed to render the

taste of food more agreeable. Among these may be noticed (see Case 44) oil of bitter almonds, vanilla, lemon, orange, and citron peel.

One of the most interesting discoveries of modern chemistry is the nature of those essences which give the various flavours to fruits. These “artificial fruit essences” have been so skilfully imitated in the laboratory of the chemist that they are extensively employed to flavour confectionery, &c. Examples of these artificially prepared fruit essences, may be seen in the case.

ACIDS.

Many of the organic acids resemble closely in their composition starch and sugar, and may to a certain extent act on the system in the same way. In the classification adopted in the Food Collection, they are classed under *medicinal or auxiliary food*.

Acetic Acid or Vinegar is obtained either from the oxidation of alcohol in fermented liquors, or from the distillation of wood. Common vinegar is obtained from the oxidation of the fermented wort of malt. A series of preparations illustrating the processes undergone in the formation of vinegar from malt has been presented to the collection by Messrs. Beaufoy and Co.

Citric Acid is contained in many fruits, but exists in greatest abundance and purity in the fruits of the orange, the lemon, the citron, the shaddock, the pommeloe, the lime, and others. All these fruits contain citric acid, and varying proportions of sugar. Citric acid can be separated from the juice of these plants in a crystalline form.

Tartaric Acid is found in the juice of the fruits of the vine tribe, more especially of the common vine. This acid gives the acidity to the fruit of the grape, and is the acid present in wines.

Malic Acid is contained in the fruits of the natural order Rosaceæ. It has the same general properties as the other acids, and is contained alone in apples and pears; whilst in cherries, plums, &c. it is mixed with other acids.

Oxalic Acid is contained in the wood sorrel, also in the common sorrel, and various species of rhubarb. Species of the latter genus are extensively cultivated in this country, and the stalks of their large leaves cut up and made into pies, puddings, &c. They are ready for use early in the spring, and are an excellent substitute for fruit in pies and tarts at that season of the year. Although oxalic acid is a *poison* when taken in considerable quantities, as ordinarily consumed it probably acts in the same way on the system as other acids.

For examples of Acids, Pickles of various kinds, &c., see Case 23.

ALKALOIDS ACTING AS STIMULANTS OR SEDATIVES.

The next and last group is that of Food Substances containing alkaloids, which act upon the nervous system as stimulants or sedatives. The principal examples are Tea, Coffee, Cocoa, Tobacco, Hemp, and Opium.

Tea and Coffee have hardly any other properties in common than the possession of an alkaloid called *Theine* or *Caffeine*, which is identical in the two. Chocolate contains a peculiar alkaloid called *Theobromine*. Paraguay tea or “maté” is the only other substance extensively used as a dietetic infusion that contains *theine*; in South America it occupies the same position in domestic economy as Chinese tea does in this country. Case 46 exhibits the chemical analysis of a pound of ordinary good Chinese Tea; and in Cases 47, 48, and 99, contiguously placed, are numerous samples of tea from China, Japan, Java, Brazil, and East India.

The analysis of a pound of Coffee is shown in Case 50; and many samples of raw coffee from various parts of the world are exhibited. The visitor’s notice is directed to the printed descriptive label concerning Coffee, which is suspended near the case containing the analysis.

Cocoa is represented by the analysis of a pound of Cocoa paste in Case 53, and by a series of the fruit pods containing the seed or nuts, presented by Messrs. Fry and Son, of Bristol, as well as by various preparations and confections. The distinguishing feature of its composition consists in the large quantities of fat and albumen which it contains; so that Cocoa not only acts as an alternative through its *theobromine*, but as a force-producing and

flesh and force-producing food. Samples of the commercial varieties of the nuts or seeds from various countries are exhibited.

TEA AND COFFEE SUBSTITUTES.

A variety of substances are exhibited in Case **49** in illustration of the use of the leaves of various plants for making potable infusions. The most important is Paraguay tea or “maté,” which contains the same alkaloid called *theine*, as the Chinese tea plant. A quantity of “maté” is shown, with calabash or dried gourd for making the infusion, and sucking pipe for drinking it. A large number of substances have been employed from time to time as substitutes for Coffee, and prepared in the same way. Many of them are shown in Case **52**. In this country none of these so-called Coffee Substitutes have established themselves in public reputation, and are seldom sold. Some of them, however, are used as adulterants of Coffee.

TOBACCO.

Although Tobacco has been only comparatively recently introduced amongst the inhabitants of the Old World, it is more extensively employed than any other narcotic. It is the produce of various species of the genus *Nicotiana*. The practice of smoking the leaves of these plants was introduced from the New World. The species, which is a native of America, and which supplies the greater proportion of the Tobacco smoked in Europe, is the *Nicotiana Tabacum*. The leaves of these plants contain an active and highly poisonous principle called *Nicotine*, which is the agent that produces the narcotic effect experienced in smoking. This narcotic effect resembles in some measure that of alcohol. Tobacco has, however, a less stimulant effect than alcohol, and produces, especially at first, a greater derangement of the general nervous system.

The chemical analysis of one pound of Tobacco is exhibited in Case **98**. Cases **93** to **97** contain a series of specimens of the leaves of species and varieties of Tobacco cultivated in various parts of the world, with samples also of the Cigars and Tobaccos manufactured from them.

Examples of varieties of Snuffs used in this and other countries will be seen in Case **97**. Snuffs are usually made from the stalks and ribs of tobacco

leaves. Tobacco is liable to be adulterated with the leaves of other plants; these, however, can be detected under the microscope. Specimens of Tobacco adulterated with other leaves are illustrated in the diagram near the cases.

Opium is used extensively as a medicine, on account of its power of alleviating pain and inducing sleep. It is a very powerful, and consequently dangerous narcotic poison, and should never be taken except under medical advice. In small doses it acts as a stimulant. On account of this latter property and its subsequent soothing influence, it has been indulged in by man, and is consumed largely in China and other parts of the world as a dietetical luxury. When taken for this purpose it is smoked, and is generally consumed with tobacco or some other leaf in a pipe. Pipes used for this purpose in China are exhibited in the cases containing Chinese food, with a collection of Chinese tobaccos, most of which appear to contain Opium.

The practice of “opium eating,” as it is called, exerts a most prejudicial effect upon the system; and although not rapidly destroying life, the victim of this habit is after a time rendered perfectly miserable if not able to procure this indulgence. As is the case with alcohol and tobacco, the system becomes accustomed to the use of this narcotic, and prodigious quantities have been consumed by those who have addicted themselves to the practice of taking it for the sake of its effect on the system.

A series of preparations from Opium, presented by the Society of Apothecaries, Apothecaries Hall, and specimens of other narcotic agents are exhibited in Case **92**. The fruits of narcotic plants are sometimes consumed in this country in mistake for other plants, producing fatal effects. Examples of some of these poisonous plants are exhibited in the Collection, as well as diagrams of others.

Before closing this necessarily brief Guide to the Food Collection, it may be as well to notice some illustrations which may be deemed exceptional to the Classification.

Fungi.—The group of Fungi contains a number of plants which are eaten as food, whilst many of them act as virulent poisons. Those which are edible contain varying quantities of starch, sugar, woody fibre, and albumen,

together with an acid called *fungic* acid. A large number of species are eaten on the continent of Europe, which are not used in this country at all. Models of the common Mushroom, the Morel, and the Truffle, are shown, as well as preserved specimens.

The visitor should notice the large Truffle from Australia called “Native Bread.” It is in Case **34**.

A series of coloured diagrams or drawings of British Fungi taken from living specimens, both edible and poisonous, are also exhibited, having been purchased from the artist, H. Worthington Smith, Esq., F.L.S.

National Foods.—Various food products of foreign nations are included in the Collection, and for the sake of illustrating *National Food* have been kept separate. It is hoped that other collections of the same kind, illustrating *national peculiarities of diet* may be formed and presented to the Museum.

In Cases **72** to **78** is an important collection of Chinese Food, which was received in 1859. The collection was procured from two districts, Shanghae and Foo-Chow-Foo, and contains many curious and interesting edible substances and preparations, some of them new to this country. Descriptive labels are attached to every specimen.

Japanese, Siamese, and East Indian food products are shown in Cases **81** to **86**.

In Case **79** will be noticed a series of *edible birds' nests* from China, Siam, Java, Borneo, and other countries. These nests are formed by swallows, two specimen of which are shown in the case.

ECONOMIC ENTOMOLOGY APPLIED TO FOOD AND FORESTRY.

This Collection, formed and arranged for the Department of Science and Art by Andrew Murray, Esq., F.L.S., is exhibited at the eastern end of the gallery, partly on the wall, and partly in glazed cases on the counter. It is intended to illustrate the ravages of such insects as are known to be destructive to alimentary substances, or that are noxious or injurious to man and domestic animals. Also, those insects known to be destructive to timber, or otherwise exemplifying the science of Entomology in its relation

to Forestry. The Collection is contained in **42** cases, to which belong **31** framed diagrams and drawings. This Entomological Collection is yet incomplete, and may be considered as still in course of formation.

ADULTERATION OF FOOD.

The extensive employment of various substances for the Adulteration of Food has led to the formation of a collection of those more commonly employed. In Cases **97** and **98** these substances are arranged according as they have been obtained from the animal, vegetable, or mineral kingdom. They have been selected principally from the results obtained by Dr. Hassall, and made known in his work "On the Adulteration of Food."

The adulteration of *Milk* with *Water* is shown in Case **54**. These examples are renewed weekly; and the use of the *lactometer* or *milk measurer*, is also exemplified in the same case. The goodness of milk can also be ascertained by means of the microscope. The diagrams on the wall represent good cream, good pure milk, bad milk, and the curd of milk as seen under the microscope.

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[4018.—5000.—7/72.]

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CATALOGUE of the COLLECTION lent by SIR RICHARD WALLACE,
BART. *Sixpence.*

ALPHABETICAL INVENTORY of the FOOD COLLECTION.
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SCIENCE DIRECTORY, with Regulations for Establishing and
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ART DIRECTORY, with Regulations for promoting Instruction in Art.
Sixpence.

* * *For information as to instruction in Science and Art in the Schools
established in the Eastern and North-eastern districts of London, see the
end cover.*

Instruction in Science and Art in the Eastern and North-Eastern Districts of the Metropolis, in connexion with the Science and Art Department.

1. Metropolitan District Schools of Art are established at the following places in the Eastern and North-eastern Districts:—

St. Thomas' Charterhouse, Goswell Street Road.

Spitalfields, 12, White Lion Street, Norton Folgate.

North London, Sandringham Road, Kingsland.

These Schools are open in the evening from 7 to 9. There are female classes at each school. Applications for admission, prospectuses, or any other information to be made at the schools in each district.

2. Night classes for instruction in drawing in connexion with the Science and Art Department are held in the following schools:—

Bethnal Green National School.

Hackney Parochial School.

Mile End, Church Street School.

Poplar, All Saints' National School.

„ St. Saviour's National School.

Stepney, St. Paul's School.

„ St. Philip's National School.

Whitechapel, St. Paul's National School.

Science classes, in various branches of science, also in connexion with the Department have been formed in the following schools:—

Bethnal Green	Birkbeck School. Abbey Street School. National School. St. James the Less, National School.
Bow	Baptist Chapel School. (Bow North) Old Ford Road School.
Hackney	Parochial School. St. Thomas Square School.
Kingsland	North London School of Art, Sandringham Road.
Mile End	Church Street School. Church of England Young Men's Society.
Poplar	St. Saviour's National School. St. Mary's School. Boys' National School. (Poplar Bromley) William Street.
Stepney	Colet Boys' School. Blue Coat School. St. Paul's School. St. Philip's National School.

Information as to fees, &c. can be obtained on application at these schools.

4. The *Science Directory* and *The Art Directory*, price 6d. each, published by the Science and Art Department, contain full particulars of the regulations for establishing and conducting Science and Art schools and classes, with lists of those in operation. To be had at the Catalogue Stall.

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The above arrangements are similar to those of the South Kensington Museum.

FOOTNOTES.

[7] The whole of the official correspondence on the subject of the establishment of this Branch Museum has been printed as a Parliamentary Paper No. 218, session of 1872.

[9] *Advantage has been taken to incorporate in this Guide much of the information contained in the Guide to the Collection (now out of print), compiled by Dr. Lankester, in 1863, for the Science and Art Department.*

*** END OF THE PROJECT GUTENBERG EBOOK A BRIEF GUIDE
TO THE FOOD COLLECTION ***

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